

■ COMPLETE NOTES - TOPIC 16

CSS Grid Layout

Two-dimensional layouts, tracks, areas, placement aur responsive grids -
Basic se Advanced tak

Web Development Course - Shauray

display grid

rows / columns

tracks

fr unit

repeat()

minmax()

auto-fit

grid areas

line placement

alignment

implicit grid

subgrid

CSS Grid Kya Hai?

Two-dimensional layout system for rows and columns

SIMPLE DEFINITION

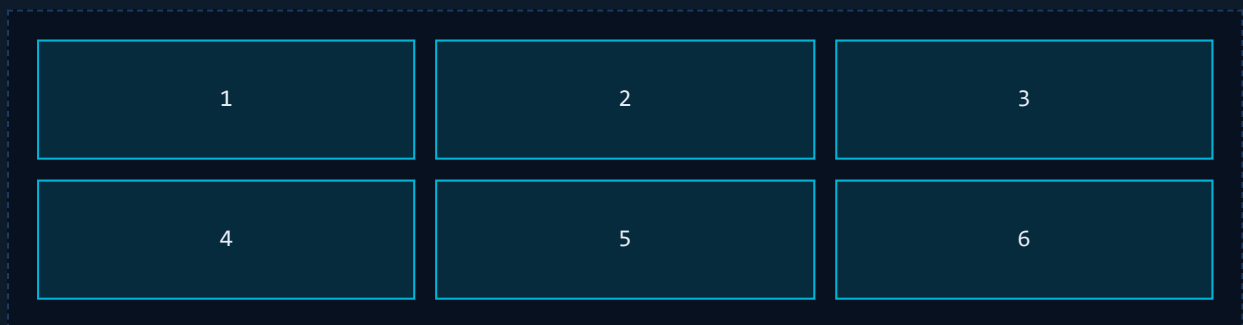
CSS Grid ek powerful layout system hai jo rows aur columns dono ko ek saath control karta hai. Flexbox mostly one-dimensional layout ke liye best hai, jabki Grid two-dimensional page/layout structure ke liye best hai.

Grid se card galleries, dashboards, page shells, image layouts, magazine layouts, forms, product listings aur complex responsive sections clean CSS se ban sakte hain. Grid parent par `display: grid` lagta hai aur uske direct children grid items ban jaate hain.

first grid

```
.grid {  
  display: grid;  
  grid-template-columns: repeat(3, 1fr);  
  gap: 1rem;  
}
```

BASIC 3 COLUMN GRID



■ Grid Vocabulary

Term	Meaning
Grid Container	Parent element jisme display: grid hota hai
Grid Item	Grid container ka direct child
Grid Line	Rows/columns ko divide karne wali numbered lines
Grid Track	Do grid lines ke beech row ya column space
Grid Cell	One row and one column ka intersection
Grid Area	Multiple cells ka named or placed region

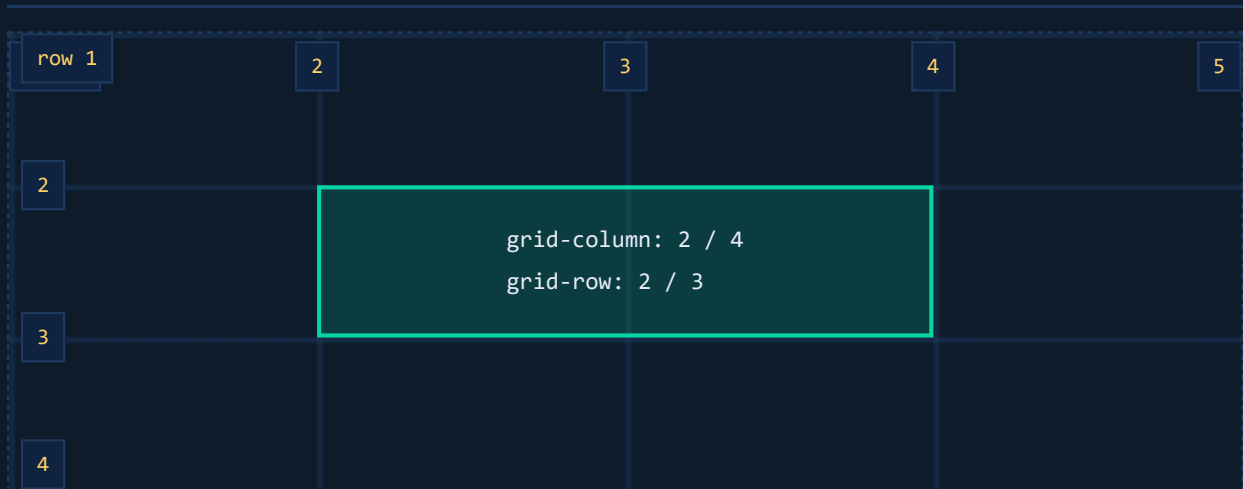
Grid Lines, Tracks and Cells

Grid ko line numbers aur tracks ke through samajhna

GRAPH CONCEPT

Grid layout invisible lines se banta hai. 4 columns ke liye 5 column lines hoti hain. 3 rows ke liye 4 row lines hoti hain. Items ko line numbers ke beech place kiya ja sakta hai.

GRID LINE GRAPH



● ● ● line placement

```
.featured {  
  grid-column: 2 / 4;  
  grid-row: 2 / 3;  
}
```

■ **Line Rule:** Grid line numbers gaps nahi, lines count karte hain. 3 columns ka matlab line 1, 2, 3, 4.

grid-template-columns and rows

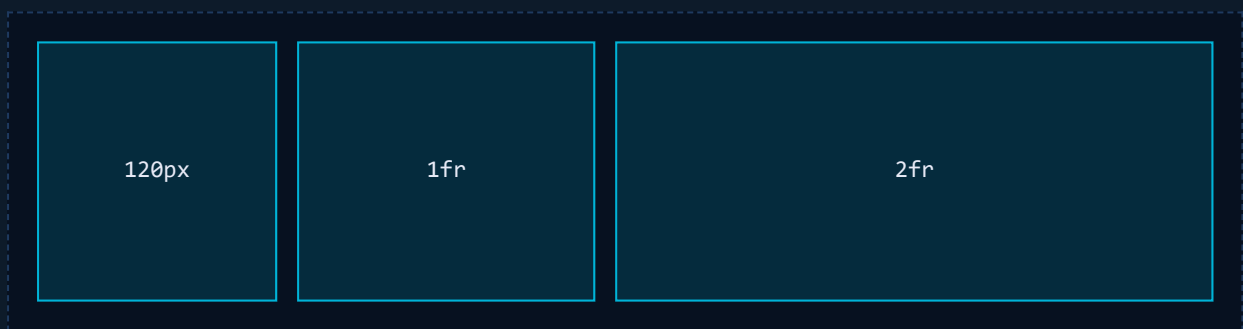
Explicit grid tracks define karna

`grid-template-columns` columns define karta hai. `grid-template-rows` rows define karta hai. In values mein px, %, rem, auto, fr, minmax(), repeat() use ho sakte hain.

columns and rows

```
.layout {  
  display: grid;  
  grid-template-columns: 240px 1fr 2fr;  
  grid-template-rows: 80px 1fr;  
  gap: 16px;  
}
```

TRACK SIZE GRAPH - 120PX 1FR 2FR



■ Track Units

Unit	Meaning	Use
px	Fixed size	Fixed sidebar, known dimensions
%	Parent size ka percentage	Relative tracks
fr	Available free space ka fraction	Flexible columns
auto	Content-based size	Natural width/height tracks
minmax()	Minimum and maximum range	Responsive cards

fr, repeat() and minmax()

Grid ka responsive sizing toolkit

IMPORTANT TOOLS

`fr` available free space ko fractions mein divide karta hai. `repeat()` repeated tracks ko short banata hai. `minmax()` track ko flexible min/max range deta hai.

fr repeat minmax

```
.grid {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
}

.responsive-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
}
```

Expression	Meaning
1fr	Available space ka 1 fraction
2fr	1fr se double share
repeat(4, 1fr)	4 equal flexible columns
minmax(200px, 1fr)	Minimum 200px, max available flexible space
repeat(auto-fit, minmax(220px, 1fr))	Responsive columns without many media queries

■ **Power Pattern:** `repeat(auto-fit, minmax(250px, 1fr))` responsive card grids ke liye top-level professional pattern hai.

gap, grid-column and grid-row

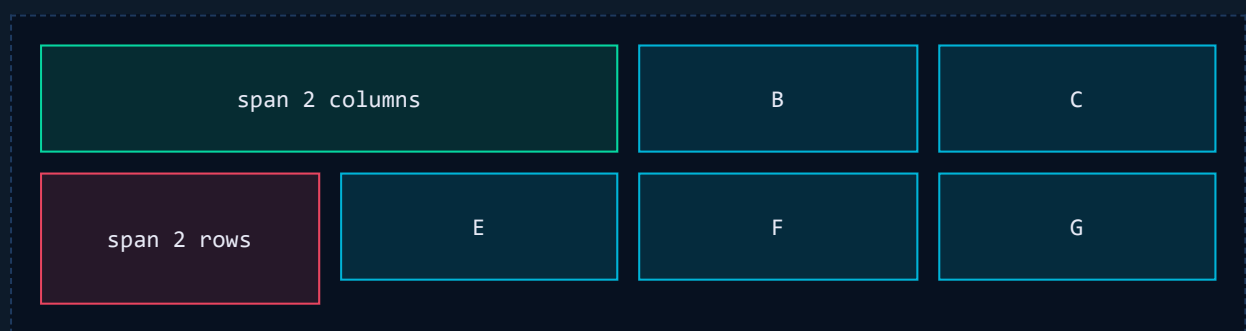
Spacing aur item spanning control karna

`gap` grid tracks ke beech spacing add karta hai. `grid-column` aur `grid-row` items ko lines ke beech place ya span karte hain.

span examples

```
.wide {  
  grid-column: span 2;  
}  
  
.tall {  
  grid-row: span 2;  
}  
  
.hero {  
  grid-column: 1 / -1;  
}
```

SPANNING GRAPH



■ **Shortcut:** `grid-column: 1 / -1` ka matlab first line se last line tak full row span karna.

grid-template-areas

Layout ko visual names ke saath define karna

DEFINITION

grid-template-areas layout ko readable names se define karta hai. Isse header, sidebar, main, footer jaisi page structures very clear CSS mein likhi ja sakti hain.

grid areas

```
.page-layout {  
  display: grid;  
  grid-template-columns: 240px 1fr;  
  grid-template-areas:  
    "header header"  
    "sidebar main"  
    "footer footer";  
}  
  
header { grid-area: header; }  
aside { grid-area: sidebar; }  
main { grid-area: main; }  
footer { grid-area: footer; }
```

TEMPLATE AREAS GRAPH



■ **Area Rule:** Same named area rectangle shape mein honi chahiye. Broken/L shaped area invalid ho sakti hai.

auto-fit vs auto-fill

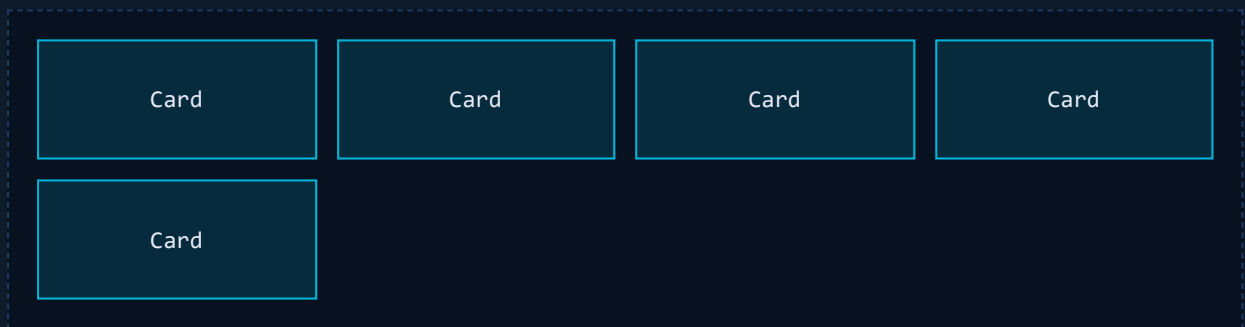
Responsive grids without exact breakpoint count

`auto-fit` aur `auto-fill` `repeat()` ke saath dynamic columns create karte hain. Dono available space ke according columns fit karte hain, but empty tracks handle karne ka behavior different hota hai.

● ● ● auto responsive grid

```
.cards {  
  display: grid;  
  grid-template-columns: repeat(auto-fit, minmax(180px, 1fr));  
  gap: 1rem;  
}
```

AUTO-FIT + MINMAX GRAPH



Value	Simple Meaning	Practical Use
<code>auto-fit</code>	Empty tracks collapse, items stretch	Most responsive card grids
<code>auto-fill</code>	Empty tracks preserved	When track slots should remain

■ **Beginner Choice:** Responsive cards ke liye usually `auto-fit` use karo. Jab empty columns preserve karne ka specific reason ho tab `auto-fill`.

Explicit and Implicit Grid

Defined tracks aur automatically created tracks

DEFINITION

Explicit grid woh hoti hai jo hum `grid-template-columns` aur `grid-template-rows` se define karte hain. Agar extra items ya placement defined grid se bahar chala jaaye, browser implicit tracks create karta hai.

● ● ● implicit rows

```
.grid {  
  display: grid;  
  grid-template-columns: repeat(3, 1fr);  
  grid-auto-rows: 120px;  
  grid-auto-flow: row;  
}
```

Property	Use
<code>grid-auto-rows</code>	Implicit rows ka size define karta hai
<code>grid-auto-columns</code>	Implicit columns ka size define karta hai
<code>grid-auto-flow</code>	Auto placement row, column, dense behavior control karta hai

■ **Advanced Note:** `grid-auto-flow: dense` gaps fill kar sakta hai, but visual order change ho sakta hai. Accessibility and reading order carefully test karo.

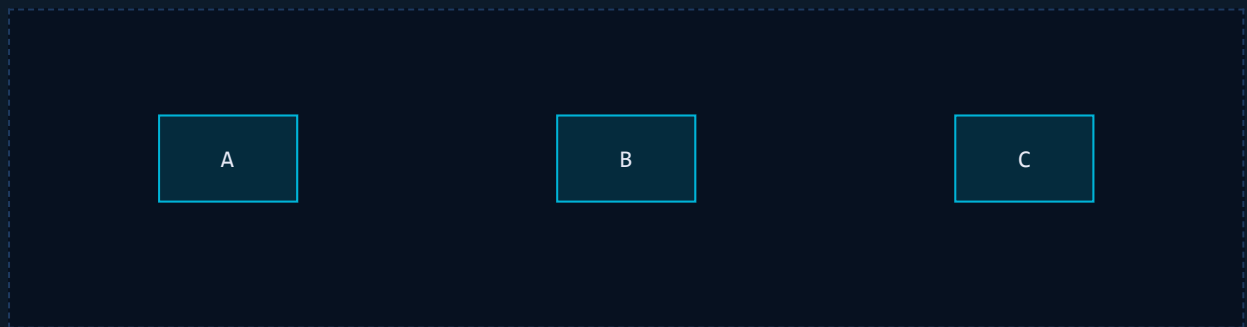
Grid Alignment

Items aur complete grid ko align karna

Grid mein alignment do levels par hoti hai: grid items apne cells ke andar kaise align honge, aur complete grid container ke available space mein kaise align hoga.

Property	Axis / Level	Use
<code>justify-items</code>	Inline axis, items inside cells	All items horizontal cell alignment
<code>align-items</code>	Block axis, items inside cells	All items vertical cell alignment
<code>place-items</code>	<code>align-items</code> + <code>justify-items</code>	Quick item centering
<code>justify-content</code>	Whole grid inline axis	Grid tracks container mein align
<code>align-content</code>	Whole grid block axis	Grid tracks vertical alignment
<code>place-content</code>	<code>align-content</code> + <code>justify-content</code>	Whole grid quick alignment

PLACE-ITEMS CENTER GRAPH



● ● ● grid alignment

```
.grid {  
  display: grid;  
  place-items: center;  
}
```

Real Project Grid Patterns

Cards, dashboard, page layout aur magazine style sections

■ Responsive Card Grid

● ● ● card grid pattern

```
.cards {  
  display: grid;  
  grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));  
  gap: 1rem;  
}
```

■ Dashboard Layout

DASHBOARD GRID GRAPH



```
.dashboard {  
  display: grid;  
  grid-template-columns: 220px 1fr 1fr;  
  grid-template-areas:  
    "nav nav nav"  
    "side chart stats"  
    "side table table";  
}
```

Grid vs Flexbox

Kaunsa layout system kab choose karen

SIMPLE DECISION

Flexbox one-dimensional flow ke liye best hai: row ya column. Grid two-dimensional structure ke liye best hai: rows and columns together. Dono competitors nahi, partners hain.

Need	Choose	Reason
Navbar links	Flexbox	One row alignment
Button group	Flexbox	Inline distribution
Product card grid	Grid	Rows and columns
Dashboard page	Grid	Named areas and 2D control
Card internal layout	Flexbox	Vertical item spacing/button bottom
Outer page shell	Grid	Header/sidebar/main/footer structure

■ **Professional Pattern:** Outer layout Grid se banao, component ke andar alignment Flexbox se. Ye combination very common hai.

Advanced Grid Concepts

Named lines, fit-content, masonry thinking, subgrid and dense flow

■ Named Grid Lines

named lines

```
.layout {  
  display: grid;  
  grid-template-columns: [sidebar-start] 240px [sidebar-end content-start] 1fr  
  [content-end];  
}  
  
.main {  
  grid-column: content-start / content-end;  
}
```

■ fit-content and clamp-like Sizing

fit content

```
.grid {  
  grid-template-columns: fit-content(260px) 1fr;  
}
```

■ Subgrid

`subgrid` child grid ko parent grid ke tracks inherit karne deta hai. Yeh advanced layout alignment ke liye useful hai, especially cards/forms mein jahan nested content parent columns align kare.

```
.card {  
  display: grid;  
  grid-template-rows: subgrid;  
}
```

■ **Advanced Note:** Subgrid modern feature hai. Use karte time browser support check karna good habit hai.

Common Mistakes and Fixes

Grid bugs ko quickly identify karna

MISTAKE

Grid items direct children rule bhoolna.
Line numbers tracks samajh lena.
fr unit ko percentage samajhna.
grid-template-areas mein broken shape banana.
Fixed columns se mobile overflow karna.
Flexbox ka kaam Grid se force karna.
align-items vs align-content confuse karna.

FIX

Direct children inspect karo.
Lines always tracks se one extra hoti hain.
fr available free space share karta hai.
Areas rectangle shape mein rakho.
minmax and auto-fit responsive pattern use karo.
One-dimensional layouts ke liye Flexbox use karo.
Item alignment and grid alignment separate samjho.

■ Debugging Checklist

Check	Question
Container	Kya parent par display: grid laga hai?
Direct child	Kya item grid container ka direct child hai?
Tracks	Columns/rows expected size define hain?
Lines	grid-column line numbers correct hain?
Areas	Template area names match kar rahe hain?
Responsive	Small screen par minmax/auto-fit ya media query needed hai?
Overflow	Any fixed width track page ko side-scroll kara raha hai?

■ **Big Rule:** Grid ko tab use karo jab rows and columns ka structure important ho. Simple single-axis alignment ke liye Flexbox zyada natural ho sakta hai.

CSS Grid - Complete Summary

Revision ke liye all important Grid properties ek jagah

■ Container Properties

Property	Meaning	Example
<code>display</code>	Grid container banana	<code>display: grid</code>
<code>grid-template-columns</code>	Explicit columns define	<code>repeat(3, 1fr)</code>
<code>grid-template-rows</code>	Explicit rows define	<code>80px 1fr auto</code>
<code>gap</code>	Rows/columns ke beech spacing	<code>gap: 1rem</code>
<code>grid-template-areas</code>	Named layout areas	<code>"header header" "side main"</code>
<code>grid-auto-rows</code>	Implicit row size	<code>grid-auto-rows: 120px</code>
<code>grid-auto-flow</code>	Auto placement direction	<code>row, column, dense</code>

■ Item Properties

Property	Meaning	Example
<code>grid-column</code>	Column placement/span	<code>1 / -1, span 2</code>
<code>grid-row</code>	Row placement/span	<code>2 / 4, span 2</code>
<code>grid-area</code>	Named area assign or shorthand	<code>grid-area: main</code>
<code>justify-self</code>	One item inline alignment	<code>center, end, stretch</code>
<code>align-self</code>	One item block alignment	<code>center, start, end</code>
<code>place-self</code>	<code>align-self</code> + <code>justify-self</code>	<code>place-self: center</code>

■ Fast Decision Guide

Need	Choose
Equal 3 columns	repeat(3, 1fr)
Responsive cards without many media queries	repeat(auto-fit, minmax(240px, 1fr))
Full width item	grid-column: 1 / -1
Item 2 columns wide	grid-column: span 2
Page shell with header/sidebar/main/footer	grid-template-areas
One row navbar	Flexbox
Complex dashboard	CSS Grid

■ Best Practices - Yaad Rakhna

DO

Rows and columns ke liye Grid use karo.
Responsive cards ke liye auto-fit + minmax use karo.
Readable page layouts ke liye grid-template-areas use karo.
Spacing ke liye gap use karo.
Line numbers carefully count karo.
Outer layout Grid, inner alignment Flexbox use karo.
Mobile overflow test karo.

AVOID

Simple one-row alignment Grid se overcomplicate karna.
Fixed widths se responsive break karna.
Broken template areas banana.
Direct child rule bhoolna.
auto-fit/auto-fill difference ignore karna.
dense flow se accessibility order break karna.
Grid and Flexbox ko competitors samajhna.

> NEXT TOPICS TO COVER

Grid vs Flexbox

Responsive Design

Container Queries

Pseudo Classes

Transitions

Animations

Forms Styling

CSS Architecture

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